Mingyang Ren

Ph.D. Candidate, University of Chinese Academy of Sciences
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Research Interests

Analysis of High-Dimensional Data, Heterogeneity Analysis, Robust Statistics, Biostatistics.

Educations

Sep. 2017 – Current
Ph.D. Candidate in Statistics
School of Mathematics Sciences, University of Chinese Academy of Sciences.
Advisor: Prof. Sanguo Zhang

Sep. 2013 – Jun. 2017
B.S. in Mathematics
School of Mathematics and Statistics, Wuhan University.

Professional Experiences

Dec. 2019 – Dec. 2020
Visiting scholar
Department of Biostatistics, Yale School of Public Health.
Mentor: Prof. Shuangge Ma

Journal Publications

- **Ren, M.**, Zhang, S., Zhang, Q., & Ma, S. (2021+). HeteroGGM: an R package for Gaussian graphical model-based heterogeneity analysis. *Bioinformatics*, https://doi.org/10.1093/bioinformatics/btab134.
- **Ren, M.**, Zhang, S., Zhang, Q. & Ma, S. (2021+). Gaussian graphical model-based heterogeneity analysis via penalized fusion. *Biometrics*, https://doi.org/10.1111/biom.13426.
- **Ren, M.**, Zhang, S. & Zhang, Q. (2020+). Robust high-dimensional regression for data with anomalous responses. *Annals of the Institute of Statistical Mathematics*, https://doi.org/10.1007/s10463-020-00764-1.
- Liu, Y., **Ren, M.** & Zhang, S. (2020+). Empirical likelihood test for regression coefficients in high-dimensional partially linear models. *Journal of Systems Science and Complexity*, https://doi.org/10.1007/s11424-020-9260-3.
- Guo, M., **Ren, M.**, Li, S. & Zhang, S. (2020+). Robust ordinal mislabel logistic regression based on γ-divergence, *Journal of University of Chinese Academy of Sciences*, Accepted.
- Li, S., **Ren, M.**, Zhang, S. et al. (2019). Effectiveness of myopia prediction model in screening children and teenager myopia, *Chinese Journal of Experimental Ophthalmology*, 37, 4: 269-273.

Preprints

- **Ren, M.**, Zhang, S., Ma, S. & Zhang, Q. Gene-environment interaction identication via penalized robust divergence. *Biometrical Journal*, Revised.
- **Ren, M.**, Zhang, Q., Zhang, S., Zhong, T., Huang, J. & Ma, S. Hierarchical cancer heterogeneity analysis based on histopathological imaging features. Under review.

Honors and Awards

Sep. 2020 **The President Scholarship of Academy of Mathematics and Systems Sciences**, **Chinese Academy of Sciences**, Excellent Prize.

2017 – 2020 **Merit Student, Outstanding Student Leader**, University of Chinese Academy of Sciences.

Jun. 2017 **Outstanding Graduate**, Wuhan University.

2013 – 2017 **Merit Student, Outstanding Student Leader, First Prize Scholarship**, Wuhan University.

Conference Reports

Nov. 2019 Identifying Gene-environment Interactions Using a Penalized Robust Divergence Approach. *The 19th Annual Conference of Chinese Association for Applied Statistics*, Beijing, China.

Oct. 2018 Robust High-Dimensional Variable Selection for Mislabel Logistic Regression. *The 11th National Annual Conference on Probability and Statistics*, Chengdu, China.

Apr. 2018 Robust Signed-rank-based High-Dimensional Test for Mean Vector. *The 4th High Dimensional Data Conference of Chinese Association for Applied Statistics*, Nanchang, China.

Invited Talks

Aug. 2019 Chinese Research Academy of Environmental Sciences, Beijing, China,

Teaching Experiences and Services

Teaching Assistant Postgraduate Courses in University of Chinese Academy of Sciences:
Advanced Mathematical Statistics (Fall 2018), Modern Statistical Methods (Spring 2019), Analysis of High-Dimensional Biostatistical Data (Summer 2019), Regression Analysis (Fall 2019).

Class Adviser 2018 Undergraduate Mathematical Class in University of Chinese Academy of Sciences.

Instructor High Dimensional Regression and Regularization
Machine Learning Seminar for undergraduates, University of Chinese Academy of Sciences (Fall 2018, Fall 2019).

Software

HeteroGGM R package (https://CRAN.R-project.org/package=HeteroGGM).